Repulsive Casimir effect between anisotropic dielectric and permeable plates

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The quantized surface mode technique was used to calculate the zero-point energy between two anisotropic parallel plates with nontrivial magnetic properties, one of which is dielectric and the other of which is permeable. The analytical approximate expressions of Casimir force and torque were obtained in the retarded limit. The repulsive Casimir effect can be acquired for some materials with suitable permittivities and permeabilities, which might be measured in a future experiment.

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I. INTRODUCTION

It is well known that the Casimir effect results from alterations of the zero-point electromagnetic energy introduced by the boundary condition [1,2]. In 1948, Casimir first pointed out that two uncharged conducting parallel plates should be attracted together by electromagnetic forces associated with the zero-point radiation field [3]. Since then, the Casimir effect has gradually become a much-discussed subject in physics [4–11]. This is due to the growing recognition of its fundamental importance in quantum field theory and also in elementary particle physics, cosmology, and condensed matter physics as well as its practical and decisive role in nanotechnology [12,13].

However, two parallel plates with nontrivial magnetic properties, one of which is a perfect conductor ($\varepsilon \to \infty$) and the other of which is an infinite permeable material ($\mu \to \infty$), can be repelled [14,15]. This setup was first proposed by Boyer who analyzed it from the viewpoint of random electrodynamics, and it is the simplest example of the repulsive Casimir force. This repulsive force is 7/8 of the magnitude of the attractive force between two perfectly conducting plates or between two infinitely permeable plates. This result can also be obtained by employing the equal time and space electromagnetic field correlators evaluated between parallel material surfaces [16]. For the general dielectric materials, Kenneth *et al.* discussed the repulsive Casimir forces between two infinite slabs with nontrivial magnetic susceptibility [17].

In a recent work [18], we employed the quantized surface mode technique to calculate the Casimir torque between two birefringent plates [19,20]. However, in order to concentrate on electric properties, we neglected the effects of the magnetic properties and assumed the magnetic permeabilities $\mu_1=\mu_2=1$ for the two plates. In this paper we will employ the same notation to calculate the retarded Casimir interaction of two general anisotropic parallel plates, especially one of which is dielectric and the other of which is permeable. We also present detailed calculations of the mechanical torque between the two parallel anisotropy plates, and this torque makes the plates rotate toward a configuration with smaller energy [21–25]. When the two plates are isotropic, we find the repulsive force can be acquired in a wide range of dielectric-magnetic materials. In particular, a repulsive force between a perfect conducting plate and an infinitely permeable one with a separation a is $F=(7/8)\pi^2\hbar c/240a^4$ per unit area. This result is consistent with that obtained in Ref. [14].

II. QUANTIZED SURFACE MODES FOR ANISOTROPIC PLATES

As shown in Fig. 1, we will consider two plates (diameter D, thickness d) made of the different uniaxial materials and kept parallel at a distance a in vacuum. The two optic axes are in the x-y plane and the z axis is orthogonal to the plates. On account of $d \gg a$ and $D \gg a$, we may consider two semi-infinite plates with in-plane optic axis as shown in Fig. 2. In the three regions the permittivity tensors are $\overleftrightarrow{\varepsilon}_1$, $\varepsilon_0=1$, and $\overleftrightarrow{\varepsilon}_2$, and the permeability tensors are $\overleftrightarrow{\mu}_1$, $\mu_0=1$, and $\overleftrightarrow{\mu}_2$, respectively.

The vacuum fluctuation fields can be determined for the system defined above. We just consider the zero-point energy associated with the surface modes which are exponentially decaying for $z<0$ and $z>a$. In general, because each region is uniform, the electric and the magnetic fields of the surface mode q can be written in the forms [18]

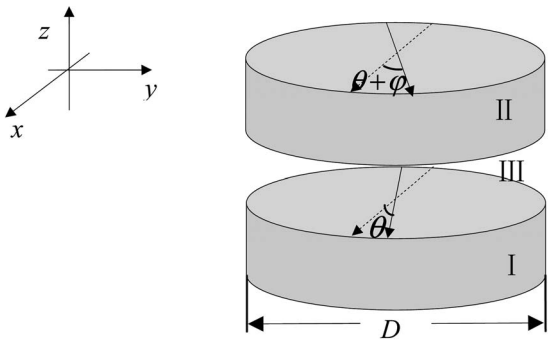


FIG. 1. Two uniaxial parallel plates with diameter D in vacuum. The optic axes of the two plates are in the x-y plane and placed in close proximity. θ and $\theta+\varphi$ are the angles between the x axis and the optic axes, with φ the angle between the two optic axes.

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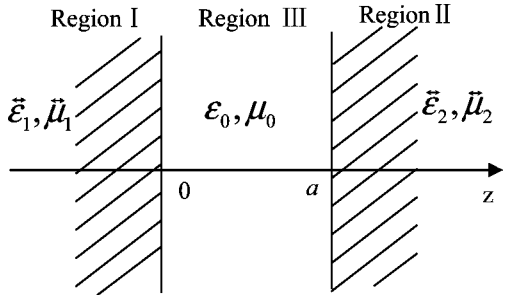


FIG. 2. The configuration of two parallel semi-infinite anisotropic plates. The optic axis of one plate is described by anisotropic permittivity tensors $\vec{\vec{\varepsilon}}_1$. The optic axis of the other plate is described by anisotropic permeability tensors $\vec{\vec{\mu}}_2$. The z axis is orthogonal to the plates.

$$\vec{E}_q = iNe^{ik_x x + ik_y y}[a_q \vec{e}_q e^{ik_z z} e^{-i\omega t} - a_q^\dagger \vec{e}_q^{\,*} e^{-ik_z z} e^{i\omega t}], \tag{1}$$

$$\vec{H}_q = Ne^{ik_x x + ik_y y}[a_q \vec{h}_q e^{ik_z z} e^{-i\omega t} + a_q^\dagger \vec{h}_q^{\,*} e^{-ik_z z} e^{i\omega t}], \tag{2}$$

where $N=(2\pi\hbar\omega/L^2)^{1/2}$ is the normalization factor, with L the length of the normalization box in the x and y directions, and $\vec{k}_\parallel=(k_x,k_y)$ is the propagation vector in the x-y plane. $a_q^\dagger$ and a_q are the usual creation and annihilation operators, respectively. The electromagnetic field polarization unit vectors $\vec{e}_q$ and $\vec{h}_q$ with frequency ω can be determined according to the classical Maxwell equations without sources in uniaxial medium,

$$\nabla \times \vec{E} = -\frac{\partial}{\partial t}\vec{\vec{\mu}} \cdot \vec{H},$$

$$\nabla \times \vec{H} = \frac{1}{c^2}\frac{\partial}{\partial t}\vec{\vec{\varepsilon}} \cdot \vec{E}. \tag{3}$$

For simplicity of the mode description, we choose the coordinates system in which $\vec{k}_\parallel$ is parallel to the x axis. Therefore the wave vector $\vec{k}$ can be expressed as

$$\vec{k} = (k_x,k_y,k_z) = K_0(\alpha,0,\gamma), \tag{4}$$

with $K_0=\omega/c$, $K_\parallel=K_0\alpha$, and $K_\perp=K_0\gamma$.

A uniaxial medium has the characteristic feature that if one of the coordinate axes is chosen to be parallel to the optic axis, the permittivity tensor of the dielectric plate and the permeability tensor of the permeable plate will be diagonal and the element associated with the optic axis is different from the remaining two diagonal ones,

$$\vec{\vec{\varepsilon}} = \varepsilon_\perp \vec{\vec{I}} + (\varepsilon_\parallel - \varepsilon_\perp)\vec{n}\vec{n}, \quad \vec{\vec{\mu}} = \mu_\perp \vec{\vec{I}} + (\mu_\parallel - \mu_\perp)\vec{n}\vec{n}, \tag{5}$$

where $\vec{\vec{I}}$ is the unit tensor and $\vec{n}$ is the unit vector in the direction of the optic axis. The subscripts $\parallel$ and $\perp$ indicate the values of the tensors along the optic axis and along the plane orthogonal to the optic axis, respectively. In our calculation, the two optic axes of two uniaxial media are both in the x-y plane and rotated by angles θ and $\theta+\varphi$ with respect to the x axis. The permittivity and permeability tensors of the two plates can be described by the following matrices:

$$\vec{\vec{\varepsilon}}_1 = \begin{pmatrix} \varepsilon_\parallel \cos^2\theta + \varepsilon_\perp \sin^2\theta & (\varepsilon_\parallel - \varepsilon_\perp)\sin\theta\cos\theta & 0 \\ (\varepsilon_\parallel - \varepsilon_\perp)\sin\theta\cos\theta & \varepsilon_\parallel \sin^2\theta + \varepsilon_\perp \cos^2\theta & 0 \\ 0 & 0 & \varepsilon_\perp \end{pmatrix}, \quad \vec{\vec{\mu}}_1 = \begin{pmatrix} \mu_1 & 0 & 0 \\ 0 & \mu_1 & 0 \\ 0 & 0 & \mu_1 \end{pmatrix},$$

$$\vec{\vec{\mu}}_2 = \begin{pmatrix} \mu_\parallel \cos^2(\theta+\varphi) + \mu_\perp \sin^2(\theta+\varphi) & (\mu_\parallel - \mu_\perp)\sin(\theta+\varphi)\cos(\theta+\varphi) & 0 \\ (\mu_\parallel - \mu_\perp)\sin(\theta+\varphi)\cos(\theta+\varphi) & \mu_\parallel \sin^2(\theta+\varphi) + \mu_\perp \cos^2(\theta+\varphi) & 0 \\ 0 & 0 & \mu_\perp \end{pmatrix},$$

$$\vec{\vec{\varepsilon}}_2 = \begin{pmatrix} \varepsilon_2 & 0 & 0 \\ 0 & \varepsilon_2 & 0 \\ 0 & 0 & \varepsilon_2 \end{pmatrix}. \tag{6}$$

Here we assume that one plate is dielectric anisotropy and the other is permeable anisotropy. The eigenequations of the transverse elements of the electromagnetic field polarization vectors $e_x, e_y, h_y, -h_x$ can be obtained by substituting $(\vec{E},\vec{H}) = (e_x, e_y, e_z, -h_x, h_y, h_z)e^{iK_0(\alpha x+\gamma z)-i\omega t}$ into Eq. (3). In the region I we can get the following result:

$$\begin{pmatrix} 0 & 0 & \mu_1 - \dfrac{\alpha^2}{\varepsilon_{33}} & 0 \\ 0 & 0 & 0 & \mu_1 \\ \varepsilon_{11} & \varepsilon_{12} & 0 & 0 \\ \varepsilon_{12} & \varepsilon_{22} - \dfrac{\alpha^2}{\mu_1} & 0 & 0 \end{pmatrix} \begin{pmatrix} e_x \\ e_y \\ ch_y \\ -ch_x \end{pmatrix} = \gamma \begin{pmatrix} e_x \\ e_y \\ ch_y \\ -ch_x \end{pmatrix}. \tag{7}$$

In order to simplify the calculation, we introduce a variable t_1 defined by $\alpha^2 = t_1^2 + \varepsilon_1\mu_1$ to characterize the parameter α. We also introduce $\varepsilon_\perp = \varepsilon_1, \varepsilon_\parallel = (1+\delta_1)\varepsilon_\perp$ and use the 4×4 matrix formulation as in Ref. [26] to calculate the wave vectors and the mode fields. The eigenvectors and the corresponding eigenvalues can be obtained in the matrix form as follows:

$$W^{\mathrm{I}} = \begin{pmatrix} -\dfrac{i\mu_1 \sin\theta}{t_1} & -\dfrac{it_1^2 \cos\theta}{\varepsilon_1 t_2} & \dfrac{i\mu_1 \sin\theta}{t_1} & \dfrac{it_1^2 \cos\theta}{\varepsilon_1 t_2} \\ \dfrac{i\mu_1 \cos\theta}{t_1} & \dfrac{i\mu_1 \sin\theta}{t_2} & -\dfrac{i\mu_1 \cos\theta}{t_1} & -\dfrac{i\mu_1 \sin\theta}{t_2} \\ \dfrac{\varepsilon_1\mu_1 \sin\theta}{t_1^2} & \cos\theta & \dfrac{\varepsilon_1\mu_1 \sin\theta}{t_1^2} & \cos\theta \\ \cos\theta & \sin\theta & \cos\theta & \sin\theta \end{pmatrix}, \tag{8}$$

$$\gamma^{\mathrm{I}} \equiv [\gamma_1\ \gamma_2\ \gamma_3\ \gamma_4] = [-it_1\ -it_2\ it_1\ it_2], \tag{9}$$

with $t_2 = \sqrt{t_1^2 + t_1^2\delta_1 \cos^2\theta - \varepsilon_1\delta_1\mu_1 \sin^2\theta}$. Each column of the matrix W^{I} represents an eigenvector. Four eigenvalues ($\gamma_1 = -it_1$, $\gamma_2 = -it_2$, $\gamma_3 = it_1$, and $\gamma_4 = it_2$) correspond to four independent mode solutions. The variables t_1 and t_2 are related with the birefringent property of the uniaxial material. For the dielectric isotropic material, we have $t_1 = t_2$ since $\delta_1 \to 0$. The general solution of the electromagnetic field should be the linear superposition of the four mode solutions, and the superposition coefficients are the amplitudes of each mode. Then the transverse elements of the electromagnetic field can be written as

$$\begin{pmatrix} E_x \\ E_y \\ cH_y \\ -cH_x \end{pmatrix} = W^{\mathrm{I}} \begin{pmatrix} e^{iK_0(\alpha x + \gamma_1 z)}A_1 \\ e^{iK_0(\alpha x + \gamma_2 z)}A_2 \\ e^{iK_0(\alpha x + \gamma_3 z)}A_3 \\ e^{iK_0(\alpha x + \gamma_4 z)}A_4 \end{pmatrix} e^{i\omega t} \tag{10}$$

in region I, where A_j is the amplitude of the jth mode solution.

The eigenequations of the transverse elements of the electromagnetic field polarization vectors $e_x, e_y, h_y, -h_x$ in region II can also be given by

$$\begin{pmatrix} 0 & 0 & \mu_2\delta_2 \sin^2(\theta+\varphi) - \dfrac{s_1^2}{\varepsilon_2} & -\mu_2\delta_2 \sin(\theta+\varphi)\cos(\theta+\varphi) \\ 0 & 0 & -\mu_2\delta_2 \sin(\theta+\varphi)\cos(\theta+\varphi) & \mu_2[1+\delta_2 \cos^2(\theta+\varphi)] \\ \varepsilon_2 & 0 & 0 & 0 \\ 0 & -\dfrac{s_1^2}{\mu_2} & 0 & 0 \end{pmatrix} \begin{pmatrix} e_x \\ e_y \\ ch_y \\ -ch_x \end{pmatrix} = \gamma \begin{pmatrix} e_x \\ e_y \\ ch_y \\ -ch_x \end{pmatrix}, \tag{11}$$

where $\alpha^2 = s_1^2 + \varepsilon_2\mu_2$ and $\mu_\perp = \mu_2, \mu_\parallel = (1+\delta_2)\mu_\perp$. Using the same method above, we can obtain the four eigenvectors and eigenvalues in the matrix form as

$$W^{\mathrm{II}} = \begin{pmatrix} \dfrac{-is_1 \cos(\theta+\varphi)}{\varepsilon_2} & -\dfrac{i\mu_2 s_2 \sin(\theta+\varphi)}{s_1^2} & \dfrac{is_1 \cos(\theta+\varphi)}{\varepsilon_2} & \dfrac{i\mu_2 s_2 \sin(\theta+\varphi)}{s_1^2} \\ \dfrac{i\mu_2 \sin(\theta+\varphi)}{s_1} & \dfrac{i\mu_2 s_2 \cos(\theta+\varphi)}{s_1^2} & -\dfrac{i\mu_2 \sin(\theta+\varphi)}{s_1} & -\dfrac{i\mu_2 s_2 \cos(\theta+\varphi)}{s_1^2} \\ \cos(\theta+\varphi) & \dfrac{\varepsilon_2\mu_2 \sin(\theta+\varphi)}{s_1^2} & \cos(\theta+\varphi) & \dfrac{\varepsilon_2\mu \sin(\theta+\varphi)}{s_1^2} \\ \sin(\theta+\varphi) & \cos(\theta+\varphi) & \sin(\theta+\varphi) & \cos(\theta+\varphi) \end{pmatrix}, \tag{12}$$

$$\gamma^{\mathrm{II}} = [-is_1\ -is_2\ is_1\ is_2], \tag{13}$$

with $s_2 = \sqrt{s_1^2 + s_1^2\delta_2 \cos^2(\theta+\varphi) - \mu_2\varepsilon_2\delta_2 \sin^2(\theta+\varphi)}$. The transverse elements of the electromagnetic field have the same form as Eq. (10), except for $W^{\mathrm{I}}, \theta, A_j$ being replaced by $W^{\mathrm{II}}, \theta+\varphi, B_j$, respectively.

Similarly, we can get the eigenvalues and eigenvectors in the vacuum region III,

$$\gamma^{\mathrm{III}} = [-it_0 \ -it_0 \ it_0 \ it_0], \tag{14}$$

$$W^{\mathrm{III}} = \begin{pmatrix} 0 & -it_0 & 0 & it_0 \\ \frac{i}{t_0} & 0 & -\frac{i}{t_0} & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \end{pmatrix}, \tag{15}$$

with $t_0=\sqrt{t_1^2+\varepsilon_1\mu_1-1}$. Here the transverse elements of the electromagnetic field mode have the same form as Eq. (10), except for $W^{\mathrm{I}}, \gamma^{\mathrm{I}}, A_j$ being replaced by $W^{\mathrm{III}}, \gamma^{\mathrm{III}}, C_j$, respectively.

Using the boundary conditions that the transverse elements of the electromagnetic field are continuous at $z=0$, a and the requirement of the surface modes $A_3=A_4=B_1=B_2=0$, we can get a system of eight linear homogeneous equations relating the unknown coefficients $A_1, A_2, B_3, B_4, C_1, C_2, C_3, C_4$. It has the nontrivial solutions on condition that the determinant of its coefficients is equal to zero, which leads to the equation for the determination of the proper frequency ω,

$$\begin{vmatrix} 1+x-cx^2\frac{\varepsilon_1}{\mu_1}\tan\theta & 1-y-dy^2s_1^2\frac{1}{\mu_2^2}\cot(\theta+\varphi) & \frac{ct_1^2x^2}{\mu_1^2} & \frac{ds_1^2y^2}{\mu_2^2} \\ 1-x+cx^2\frac{\varepsilon_1}{\mu_1}\tan\theta & \left[1+y+dy^2s_1^2\frac{1}{\mu_2^2}\cot(\theta+\varphi)\right]K & \frac{-ct_1^2x^2}{\mu_1^2} & \frac{-ds_1^2y^2K}{\mu_2^2} \\ -c & -d & 1+\frac{\mu_1}{\varepsilon_1 x}+\frac{ct_1^2\cot\theta}{\varepsilon_1\mu_1} & 1-\frac{\mu_2}{\varepsilon_2 y}+d\tan(\theta+\varphi) \\ c & dK & 1-\frac{\mu_1}{\varepsilon_1 x}-\frac{ct_1^2\cot\theta}{\varepsilon_1\mu_1} & 1+\frac{\mu_2}{\varepsilon_2 y}-d\tan(\theta+\varphi)K \end{vmatrix} = 0, \tag{16}$$

with $x=(\mu_1\sqrt{t_1^2+\varepsilon_1\mu_1-1})/t_1$, $y=(\mu_2\sqrt{s_1^2+\varepsilon_2\mu_2-1})/s_1$, and $K=e^{2K_0t_0a}$,

$$c = \left(\frac{t_1}{\sqrt{t_1^2+t_1^2\delta_1\cos^2\theta-\varepsilon_1\mu_1\delta_1\sin^2\theta}}-1\right) \times \frac{\mu_1^2}{x(t_1^2\cot\theta-\varepsilon_1\mu_1\tan\theta)}$$

$$d = \left(\frac{\sqrt{s_1^2+s_1^2\delta_2\cos^2(\theta+\varphi)-\mu\delta_2\sin^2(\theta+\varphi)}}{s_1}-1\right) \times \frac{\mu_2^2}{y[s_1^2\cot(\theta+\varphi)-\varepsilon_2\mu_2\tan(\theta+\varphi)]}.$$

We assume that $\delta_1, \delta_2 \ll 1$. From Eq. (16), after some simple but lengthy algebra we obtain the approximate result

$$(1-Q^{\perp^2})_{\omega_N^{\perp}}(1-Q^{\parallel^2})_{\omega_N^{\parallel}} = 0, \tag{17}$$

$$Q^{\perp^2} = y_1(1+a_1\delta_1+a_2\delta_2+a_3\delta_1\delta_2+a_4\delta_1^2+a_5\delta_2^2) \times e^{-(2\omega a\sqrt{t_1^2+\varepsilon_1\mu_1-1})/c}, \tag{18}$$

$$Q^{\parallel^2} = y_2(1+b_1\delta_1+b_2\delta_2+b_3\delta_1\delta_2+b_4\delta_1^2+b_5\delta_2^2) \times e^{-(2\omega a\sqrt{t_1^2+\varepsilon_1\mu_1-1})/c}, \tag{19}$$

with $y_1=[(x-1)(y-1)]/[(x+1)(y+1)]$, $y_2=[(x\varepsilon_1-\mu_1)(y\varepsilon_2-\mu_2)]/[(x\varepsilon_1+\mu_1)(y\varepsilon_2+\mu_2)]$ and the coefficients

$$a_1 = \frac{x\varepsilon_1\mu_1-x\cos(2\theta)\varepsilon_1\mu_1}{2t_1^2(x^2-1)}, \quad b_1 = \frac{x\varepsilon_1\mu_1+x\cos 2\theta\varepsilon_1\mu_1}{2(x^2\varepsilon^2{}_1-\mu^2{}_1)},$$

$$a_2 = \frac{y[1+\cos 2(\theta+\varphi)]}{2(y^2-1)}, \quad b_2 = \frac{y[1-\cos 2(\theta+\varphi)]\varepsilon_2^2\mu_2^2}{2s_1^2(y^2\varepsilon_2^2-\mu_2^2)},$$

$$a_3 = -\frac{xy\varepsilon_1\mu_1\cos 2\varphi}{8t_1^2(-1+x^2)(-1+y^2)} - \frac{y_1}{8(y_1-y_2)}\frac{\varepsilon_1\varepsilon_2(s_1^2y^2\mu_1^2+t_1^2x^2\mu_2^2)\cos 2\varphi}{s_1^2t_1^2(1+x)(1+y)(x\varepsilon_1+\mu_1)(y\varepsilon_2+\mu_2)} + C_1 + f_1(\varphi,\theta),$$

$$b_3 = -\frac{xy\varepsilon_1\varepsilon_2^2\mu_1\mu_2^2\cos 2\varphi}{8s_1^2(x^2\varepsilon_1^2-\mu_1^2)(y^2\varepsilon_2^2-\mu_2^2)} + \frac{y_2}{8(y_1-y_2)}\frac{\varepsilon_1\varepsilon_2(s_1^2y^2\mu_1^2+t_1^2x^2\mu_2^2)\cos 2\varphi}{s_1^2t_1^2(1+x)(1+y)(x\varepsilon_1+\mu_1)(y\varepsilon_2+\mu_2)} + C_2 + f_2(\varphi,\theta).$$

The expressions of C_1, $f_1(\varphi,\theta)$, C_2, and $f_2(\varphi,\theta)$ are very complicated. $f_1(\varphi,\theta)$ and $f_2(\varphi,\theta)$ are period functions of θ. In the following calculation, we will expand the Casimir force to the first order of δ_1, δ_2 and the Casimir torque to the second order of δ_1, δ_2. Since $f_1(\varphi,\theta)$, $f_2(\varphi,\theta)$, a_4, a_5, b_4, b_5 do not contribute to the last force $F(a)$ and torque $\tau(\varphi)$, here we do not write them out in the above equations.

III. CASIMIR FORCE BETWEEN ANISOTROPIC PLATES

The electromagnetic energy associated with the mode $(\vec{k},\omega_N^{\perp})$ and $(\vec{k},\omega_N^{\parallel})$ are the zero-point energy $\hbar\omega_N^{\perp}/2$ and $\hbar\omega_N^{\parallel}/2$, that is,

$$\langle 0|\frac{1}{8\pi}\int d\vec{r}\left[\frac{d(\varepsilon\omega)}{d\omega}|\vec{E}_q|^2+\frac{d(\mu\omega)}{d\omega}|\vec{H}_q|^2\right]_{\omega_N^{\perp}}|0\rangle=\frac{1}{2}\hbar\omega_N^{\perp}, \tag{20}$$

$$\langle 0|\frac{1}{8\pi}\int d\vec{r}\left[\frac{d(\varepsilon\omega)}{d\omega}|\vec{E}_q|^2+\frac{d(\mu\omega)}{d\omega}|\vec{H}_q|^2\right]_{\omega_N^{\parallel}}|0\rangle=\frac{1}{2}\hbar\omega_N^{\parallel}, \tag{21}$$

where $|0\rangle$ represents the vacuum photon state. The total energy E of the system is simply the sum of the zero-point energies of the surface modes,

$$E=\sum_{\vec{k},N}\frac{1}{2}\hbar\omega_N^{\perp}+\sum_{\vec{k},N}\frac{1}{2}\hbar\omega_N^{\parallel}. \tag{22}$$

Taking the variable change $\omega=i\xi$, and using the argument theorem as applied in Refs. [18,19] to sum over $\vec{k}$ and N, the total energy can be written as

$$E(a,\varphi)=\frac{\hbar}{8\pi^3}\int_0^{\infty}dkk\int_0^{2\pi}d\theta\int_0^{\infty}d\xi[\ln G^{\perp}(i\xi)+\ln G^{\parallel}(i\xi)], \tag{23}$$

with $G^{\perp}=1-Q^{\perp^2}$ and $G^{\parallel}=1-Q^{\parallel^2}$. Following Ref. [27] and introducing a variable p defined by $k^2=\frac{\xi^2}{c^2}(p^2-1)$, Eq. (23) may be rearranged as some known results as follows:

$$E(a,\varphi)=\frac{\hbar}{8\pi^3c^2}\int_1^{\infty}pdp\int_0^{2\pi}d\theta\int_0^{\infty}\xi^2d\xi[\ln G^{\perp}(i\xi)+\ln G^{\parallel}(i\xi)]. \tag{24}$$

At large distances $a\gg\lambda_0$ (where λ_0 is the characteristic absorption wavelength of the semispace dielectric matter), the permittivities and permeabilities can be represented by their static values at $\xi=0$. Substituting $\chi=2p\xi a/c$ into Eq. (24), we obtain

$$E(a,\varphi)=-\frac{\hbar c\pi}{30a^3}\frac{5}{32\pi^4}\int_1^{\infty}\frac{dp}{p^2}\int_0^{2\pi}d\theta\times\int_0^{\infty}\chi^3d\chi\left[\frac{1}{\frac{1}{Q^{\perp^2}}-1}+\frac{1}{\frac{1}{Q^{\parallel^2}}-1}\right], \tag{25}$$

$$F(a,\varphi)=-\frac{\hbar c\pi}{10a^4}\frac{5}{32\pi^4}\int_1^{\infty}\frac{dp}{p^2}\int_0^{2\pi}d\theta\times\int_0^{\infty}\chi^3d\chi\left[\frac{1}{\frac{1}{Q^{\perp^2}}-1}+\frac{1}{\frac{1}{Q^{\parallel^2}}-1}\right]. \tag{26}$$

If $\delta_1=0$, $\delta_2=0$, Eqs. (25) and (26) return back to the results in the isotropic case. Then the Casimir energy and force in the two semi-infinite isotropic plates will be

$$E_0(a)=-\frac{\hbar c}{96\pi^2a^3}\int_1^{\infty}\frac{dp}{p^2}\int_0^{\infty}\chi^3d\chi\left[\frac{y_1e^{-\chi}}{1-y_1e^{-\chi}}+\frac{y_2e^{-\chi}}{1-y_2e^{-\chi}}\right], \tag{27}$$

$$F_0(a)=-\frac{\hbar c}{32\pi^2a^4}\int_1^{\infty}\frac{dp}{p^2}\int_0^{\infty}\chi^3d\chi\left[\frac{y_1e^{-\chi}}{1-y_1e^{-\chi}}+\frac{y_2e^{-\chi}}{1-y_2e^{-\chi}}\right]=-\frac{3\hbar c}{16\pi^2a^4}\int_1^{\infty}\frac{dp}{p^2}[\mathrm{Li}_4(y_1)+\mathrm{Li}_4(y_2)], \tag{28}$$

where $\mathrm{Li}_4(x)=\Sigma_{n=1}^{\infty}(x^n/n^4)$ is a polylogarithmic function.

Substituting Eqs. (18) and (19) into Eq. (26), we can get the Casimir force in the anisotropic case,

$$F(a)=F_0(a)+\delta F(a), \tag{29}$$

with $\delta F(a)=-(\hbar c\pi^2/240a^4)(\Phi_1\delta_1+\Phi_2\delta_2)$ and

$$\Phi_1=\frac{15\varepsilon_1\mu_1}{4\pi^4}\int_1^{\infty}\frac{dp}{p^2}\left[\frac{x(1-y)\mathrm{Li}_3(y_1)}{(p^2+\varepsilon_1\mu_1-1)(x+1)^2(y+1)}+\frac{x(y\varepsilon_2-\mu_2)\mathrm{Li}_3(y_2)}{(x\varepsilon_1+\mu_1)^2(y\varepsilon_2+\mu_2)}\right],$$

$$\Phi_2=\frac{45}{2\pi^4}\int_1^{\infty}\frac{dp}{p^2}\left[\frac{y(x-1)\mathrm{Li}_3(y_1)}{(y+1)^2(x+1)y_1}-\frac{y(x\varepsilon_1-\mu_1)\varepsilon_2^2\mu_2^2\mathrm{Li}_3(y_2)}{(p^2+\varepsilon_1\mu_1-1)(y\varepsilon_2+\mu_2)^2(x\varepsilon_1+\mu_1)y_2}\right].$$

From above it is clear that the force between the anisotropic plates is the sum of the isotropic result F_0 and a correctional term $\delta F(a)$. Considering that the expression of the force Eq. (28) is quite complicated, we will make a simple statement regarding the direction of the force in the case of high permeability and permittivity, i.e., $Z_i=\sqrt{\mu_i/\varepsilon_i}\to$const and $(\varepsilon_i,\mu_i)\to\infty$ with $(i=1,2)$,

$$F_0(a)=-\frac{\hbar c}{32a^4}\frac{1}{\pi^2}\left\{2+\frac{2(Z_1+Z_2)}{Z_2-Z_1}\left[-\frac{1}{Z_1}\ln(1+Z_1)+\frac{1}{Z_2}\ln(1+Z_2)+Z_1\ln\left(1+\frac{1}{Z_1}\right)-Z_2\ln\left(1+\frac{1}{Z_2}\right)\right]\right\}. \tag{30}$$

Equation (30) is the function of the impedance $Z_i=\sqrt{\mu_i/\varepsilon_i}$. The border curve defined by $F_0(Z_1,Z_2)=0$ is shown in Fig. 3. We observe from the figure that whether or not the force is repulsive only depends on Z_i, and the repulsive force can be acquired in a wide range of dielectric-magnetic materials. As an example, we take the dielectric plate made of a ferroelectric substance $BaTiO_3$ ($\varepsilon_1=2000$, $\mu_1=10$) and the permeable plate made of a garnet YIG ($\varepsilon_2=14.5$, $\mu_2=8000$). When they are placed parallel in the vacuum with a distance of 1 μm, the repulsive force will be about 1.3×10^{-4} N/m^2. Although testing this repulsive force still requires a number

of auxiliary measurements of the physical properties of the materials and an extensive theoretical analysis for the effects of nonidealities, this force may be measured in the future experiment.

In addition, we can calculate the force between a perfect conductor and an infinitely permeable material. Substituting $\varepsilon_1 \to \infty, \mu_2 \to \infty$ into Eq. (28), the repulsive force will become $F_0(a) = (7/8)\pi^2 \hbar c A / 240 a^4$, which is consistent with Ref. [14].

IV. CASIMIR TORQUE ON ANISOTROPIC PLATES

The expression of the energy per unit area given by Eq. (25) can be simplified and reads

$$E(a,\varphi) \approx E_0(a) - \frac{\hbar c \pi^2}{720 a^3} \psi \delta_1 \delta_2 \cos(2\varphi). \tag{31}$$

The vacuum torque per unit area is

$$\tau(\varphi) = -\frac{\partial E(a,\varphi)}{\partial \varphi} \approx -\frac{\hbar c \pi^2}{360 a^3} \sin(2\varphi) \delta_1 \delta_2 \psi. \tag{32}$$

The relation between ψ and $\varepsilon_1, \mu_1, \varepsilon_2, \mu_2$ can be expressed as

$$\psi = \frac{15}{2\pi^4} \int_1^\infty \frac{dp}{p^2} \int_0^\infty \chi^3 d\chi \left[\frac{e^{-\chi}}{(1 - y_1 e^{-\chi})^2} \frac{xy\varepsilon_1\mu_1}{8(p^2 + \varepsilon_1\mu_1 - 1)(1+x)^2(1+y)^2} \left(1 + 2\frac{y_1 e^{-\chi}}{1 - y_1 e^{-\chi}} \right) \right.$$
$$+ \frac{e^{-\chi}}{(1 - y_2 e^{-\chi})^2} \frac{xy\varepsilon_1\mu_1\varepsilon_2^2\mu_2^2}{8(p^2 + \varepsilon_2\mu_2 - 1)(x\varepsilon_1 + \mu_1)^2(y\varepsilon_2 + \mu_2)^2} \left(1 + 2\frac{y_2 e^{-\chi}}{1 - y_2 e^{-\chi}} \right)$$
$$\left. + \frac{(y_1 + y_2 - 2y_1 y_2 e^{-\chi}) e^{-\chi}}{(1 - y_1 e^{-\chi})^2 (1 - y_2 e^{-\chi})^2} \frac{\varepsilon_1\varepsilon_2 \left(\frac{y^2\mu_1^2}{(p^2 + \varepsilon_1\mu_1 - 1)} + \frac{x^2\mu_2^2}{(p^2 + \varepsilon_2\mu_2 - 1)} \right)}{8(1+x)(1+y)(x\varepsilon_1 + \mu_1)(y\varepsilon_2 + \mu_2)} \right]. \tag{33}$$

In order to make a simple statement regarding the direction of the torque, we set the impedance $Z_i = \sqrt{\mu_i / \varepsilon_i} \to \text{const}$ for $(\varepsilon_i, \mu_i) \to \infty$. Then we can obtain the approximate result

$$\psi = \frac{45}{8\pi^4} \int_1^\infty dp \left[\frac{Z_1 Z_2}{(1 + pZ_1)^2 (1 + pZ_2)^2} + \frac{Z_1 Z_2}{(p + Z_1)^2 (p + Z_2)^2} \right.$$
$$\left. + \frac{2 Z_1 Z_2 \left(\frac{(pZ_1 - 1)(pZ_2 - 1)}{(pZ_1 + 1)(pZ_2 + 1)} + \frac{(p - Z_1)(p - Z_2)}{(p + Z_1)(p + Z_2)} \right)}{(1 + pZ_1)(1 + pZ_2)(p + Z_1)(p + Z_2)} \right]. \tag{34}$$

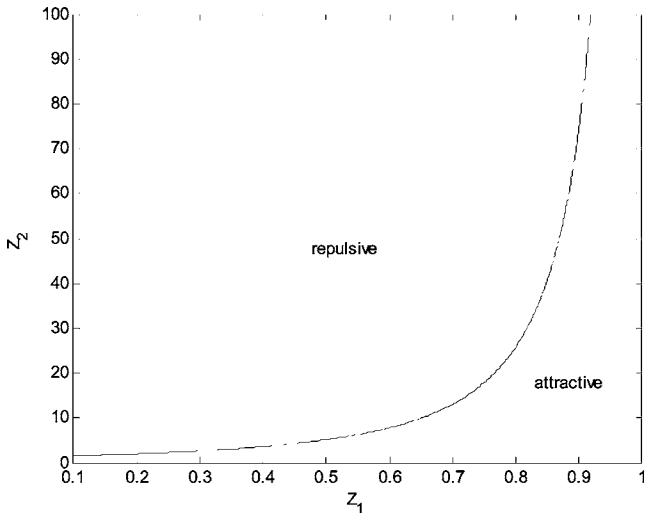


FIG. 3. The repulsive and attracting regions in the Z_1, Z_2 plane. The repulsive regions correspond with $F_0(Z_1, Z_2) > 0$, and the attracting regions correspond with $F_0(Z_1, Z_2) < 0$.

Figure 4 shows the border curve defined by $\psi(Z_1, Z_2) = 0$. From the expression of the torque given by Eq. (32), we find that the torque varies as $\sin(2\varphi)$ and has a maximum magnitude at $\varphi = \pi/4$. The direction of the torque also depends on the impedance Z_1, Z_2. The torque deriving from the anisotropy of the zero-point energy makes the two plates rotate towards the configuration with smaller energy in which their optic axes are aligned (or orthogonal). That is to say that the plates will rotate to position $\varphi = 0, \pi$ in the zone $\Psi > 0$ and rotate to position $\varphi = \pi/2$ in the zone $\Psi < 0$.

V. SUMMARY

In this paper, we have shown how to employ the quantum surface modes technique to calculate the Casimir energy between two anisotropic parallel plates, one of which is dielectric and the other of which is permeable. The analytical approximate expressions of Casimir force and torque were

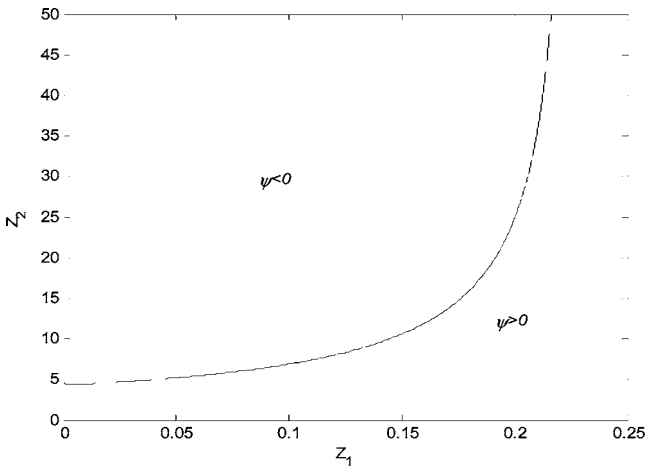


FIG. 4. The border curve of $\psi(Z_1,Z_2)=0$ in the Z_1,Z_2 plane.

obtained in the retarded limit. The repulsive Casimir effect can be acquired for some materials with suitable permittivities and permeabilities, such as ferroelectric substance ($BaTiO_3$) and garnet (YIG). In general, the algebraic sign of the Casimir force, the attractive or repulsive force, is dependent on the boundary conditions and boundary geometric configurations. For instance, in the case of a spherically conducting shell, the repulsive Casimir force can be accounted the properties of the zeros of spherical Bessel functions [28]. On the other hand, for the two parallel plates, the repulsive Casimir force between dielectric and permeable materials arises from the nontrivial electric and magnetic polarizable properties. This is similar to the repulsive Casimir-Polder force between an electric polarizable particle and a magnetic polarizable one [29]. From the point of experiment, with the development of the micronanotechnology, the repulsive Casimir effect including the force and torque might be measurable in the future.

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